Master Cross-Domain Verification Manual

Unified Multi-Physics Equation Coupling & Parameter Handshakes for Tokamak Reactor Archiving

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1. Executive Summary & Multiphysics Architecture

Next-generation high-field tokamak fusion power plants operating at toroidal magnetic fields B0 >= 10 T and plasma currents Ip >= 8 MA require a unified multiphysics engineering architecture. This manual documents the master 5-phase sequential framework linking core equilibrium magnetics, plasma-facing liquid metal hydrodynamics (Paper I), high-field REBCO superconducting magnets (Paper II), vacuum vessel 3D anisotropic structural mechanics (Paper III), and liquid metal blanket MHD flows (Paper IV).

2. Governing Mathematical Formulations & Phase Handshakes

Phase 1: Core Equilibrium Magnetics & Transport

Executes Grad-Shafranov equilibrium solvers to establish magnetic field topology B0 >= 10-20 T and parallel heat flux q_parallel.

Phase 2: PFC Hydrodynamics & Thermal CPS (Paper I)

Evaluates Taylor-Couette flow stability (Ta <= 45.8), Young-Laplace capillary retention pressure Delta P_cap = 2 gamma cos(theta) / r_p, Carman-Kozeny permeability degradation K(dpa), and critical thermal dryout heat flux q''_dryout.

Phase 3: Blanket MHD & Flow Channels (Paper IV)

Resolves 3D Navier-Stokes-Maxwell liquid metal flow in Pb-17Li channels at Hartmann number Ha ~ 10^4 (benchmark Ha = 500), boundary layer thinning delta_H = a / Ha = 0.20 mm, MHD pressure drop dP/dz = 0.45 MPa/m, and Shercliff side-wall jet velocities U_jet = 15 U0.

Phase 4: Transient Disruptions & REBCO Magnet Quench (Paper II)

Models unmitigated current quenches (I_halo = 1.8 MA) coupled to non-insulated co-wound REBCO tape stacks with turn-to-turn contact resistivity rho_ct = 15.0 uOhm.cm2, characteristic time constant tau_L/R = 1.0-3.0 s, and peak quench hot-spot temperature T_max <= 200 K.

Phase 5: Vacuum Vessel Structural Yield & Reserves (Paper III)

Evaluates Inconel 718 shear key pin interlocks under asymmetric halo current forces (DAF = 1.85) using the 3D anisotropic Tsai-Wu interactive failure criterion f_TW = 0.244 <= 1.0, establishing a Strength Reserve Ratio SRR = 1.61.

3. Master Cross-Domain Parameter Handshake Matrix

Operational State	Fluence (dpa)	Pore r_p (um)	Capillary P (kPa)	Dryout Flux (MW/m2)	Safety Factor	Tsai-Wu f_TW	Status
Pristine State	0.0	25.00	27.05	41.50	2.77	0.244	Baseline Optimal
Replacement Limit	9.8	17.54	38.55	22.50	1.50	0.244	Mandatory Replacement
Dryout Parity	13.0	16.49	41.00	15.00	1.00	0.244	Thermal Dryout Limit
Extreme Damage	20.0	15.20	44.48	8.69	0.58	0.244	Sub-critical Failure